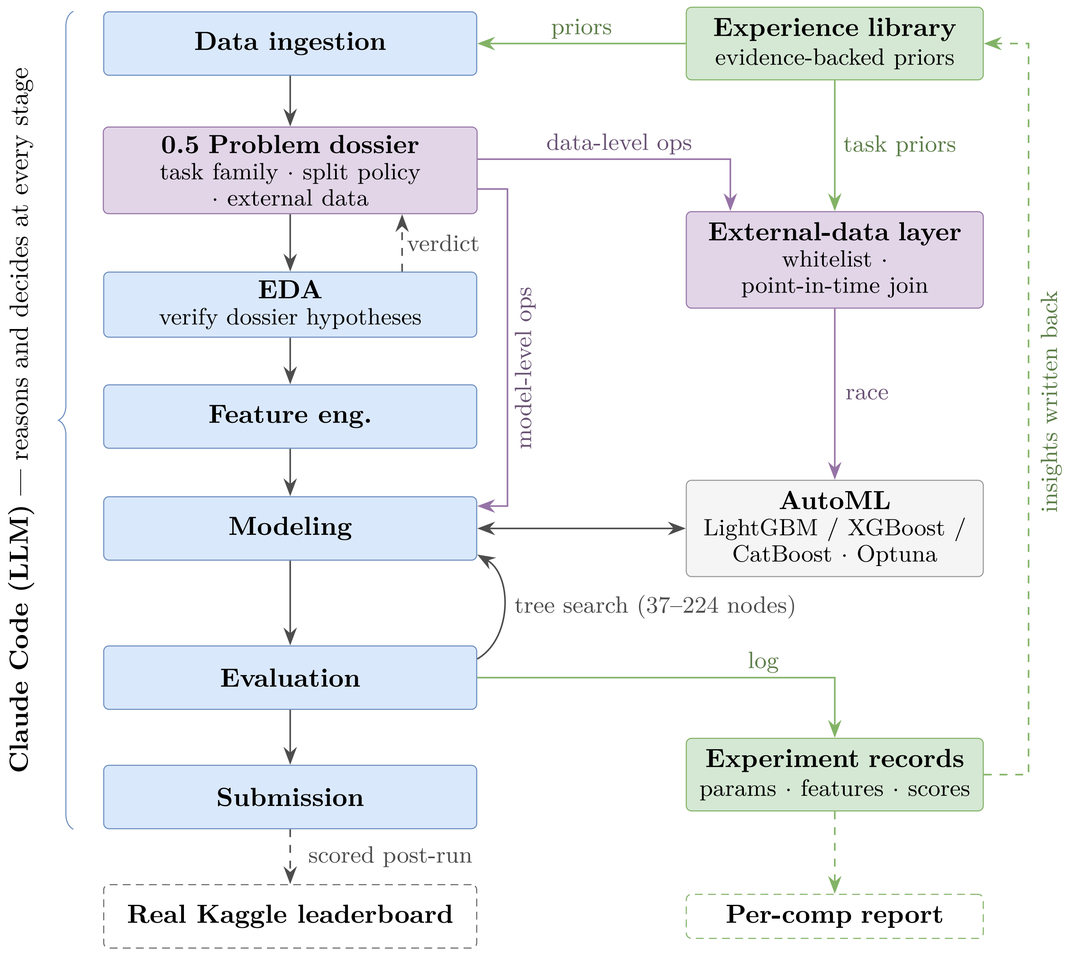
An End-to-End AI Agent for Kaggle Competitions

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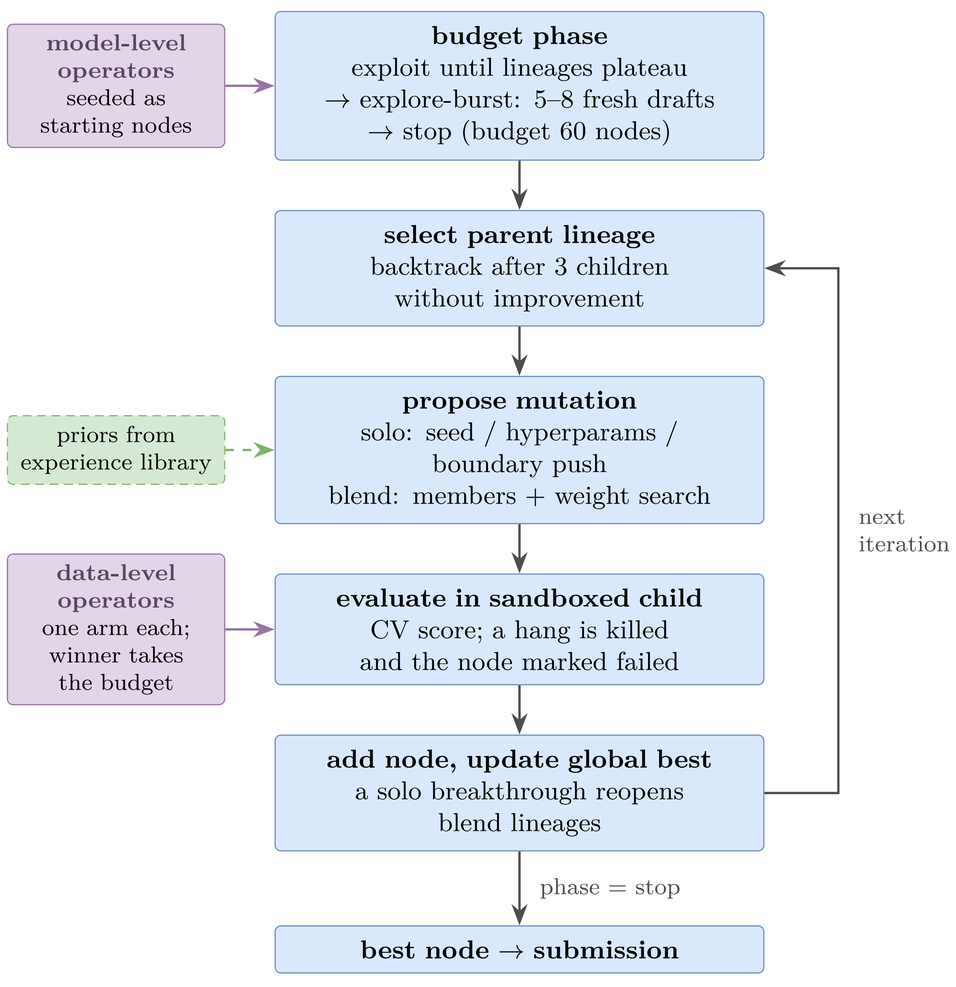
1 · Motivation

- **Challenges of Kaggle competitions**
 - **A competition requires the expert loop**
 This includes validation design, feature extraction, model selection, and ensemble building. The top entries are separated by a very small margin.
 - **Data preparation is repetitive**
 The same cleaning, split design, and drift checks are done on every dataset.
- **Why an AI agent**
 - **A competition is a *scorable* task**
 Every competitor delivers a number, so it can be optimized automatically.
 - **A competition requires tireless, systematic search**
 Every experiment is recorded and should be reproducible.
 - **Recent success of agents for Kaggle competitions**
 Notable cases include tree search over code [1], and tree search fed injected research ideas [2].

2 · Workflow



The LLM reasons and decides at every stage; the **problem dossier** classifies the task and injects external data *upstream*, before any modeling. EDA then tests the dossier's hypotheses, feature engineering and modeling follow, and evaluation returns the score the tree search uses to pick the next candidate. Purple = injection layer · green = evidence loop · gray = AutoML tools; dashed = feedback or post-run.



Tree-search loop: every node is a complete candidate solution, ensembles included. The search improves the best lineage until it stops paying off, then spends one round on 5–8 brand-new candidates before submitting the best node.

3 · Results

Our agent "Sinica" was run in audited sandboxes for all 23 competitions (20 ordinary competitions in Table 1 and 3 special competitions in Table 2). The results of the 20 ordinary competitions were scored on the real leaderboard, while the results of the 3 special competitions were scored offline with MLE-bench [3]. We compared these results with the results delivered by two other agents, AIDE [1] (tree search over code) and the NVIDIA Kaggle Agent [4] (kernel reproduction). **Among the 23 competitions, our agent "Sinica" obtains the best performance with 13 wins and 10 losses.**

Remark: MLE-bench is an offline benchmark that replays finished Kaggle competitions with their original data and metrics.

Table 1: ordinary competitions

Competition	Metric	Sinica	NVIDIA	AIDE
s3e1	RMSE ↓	**0.55275**	0.55550	0.55552
s3e3	AUC ↑	0.87157	**0.89537**	0.86863
s3e5	QWK ↑	0.57223	0.56974	**0.58138**
s3e7	AUC ↑	0.90441	**0.90458**	0.90109
s3e9	RMSE ↓	12.3319	**12.2291**	12.3067
s3e11	RMSLE ↓	**0.29368**	0.29624	0.29445
s3e14	MAE ↓	333.186	**330.716**	332.316
s3e16	MAE ↓	**1.33822**	1.34001	1.34224
s3e19	SMAPE ↓	49.4088	**48.2656**	48.3413
s4e11	Acc. ↑	**0.94107**	0.94076	0.94056
s5e1	MAPE ↓	**0.08401**	0.14594	0.17917
s5e10	RMSE ↓	**0.05576**	0.05588	0.05579
s6e1	Error ↓	**8.71516**	8.73239	8.72191
s6e2	AUC ↑	0.95505	0.95508	**0.95516**
afsis	MCRMSE ↓	**0.48917**	0.69855	0.49861
cat-in-the-dat	AUC ↑	**0.80243**	0.77084	0.80217
conway	MAE ↓	**0.10414**	0.11189	0.11065
tps-aug-2022	AUC ↑	0.58930	0.58730	**0.59106**
tps-jan-2022	SMAPE ↓	6.29895	**4.63651**	10.6664
tps-sep-2022	SMAPE ↓	28.4961	**5.48351**	9.72095
Best private score		**10 / 20**	7 / 20	3 / 20

Bold = best of the three.

Table 2: special competitions

Competition	Metric	Sinica	NVIDIA	AIDE
us-patent	Pearson r ↑	**0.86658**	0.86160	0.65821
ventilator	MAE ↓	**0.16525**	0.40093	0.34591
SIIM-ISIC	AUC ↑	**0.93529**	0.89822	0.88579

Graded by MLE-bench. **Bold** = best of the three.

4 · Disadvantages of the agent-based approach

- **The agent is strong at coding, weak at judging**
 The agent efficiently works through whatever problem it is given. However, it does not ask whether a change is worth making. This often leads to a poor design choice that yields bugs.
- **The agent can only target the local validation score**
 As a result, the local validation score can look outstanding while the real leaderboard score is mediocre.
- **The agent often buries the main question in detail**
 AI-written reports put too much focus on technical details, drifting away from what the reader wants to know.
- **The agent has bad writing habits**
 Redundant headings and reinvented terms often come back in every draft. Verifying and modifying them by hand is exhausting.

5 · Practical advice for building AI agents for data science

- **Build an isolated environment before running the agent**
 This prevents any access to the leaderboard or another run's files.
- **Justify the design before building the pipeline**
 This mitigates redesign cost, debugging time, and a full rerun of earlier competitions.
- **Inject external knowledge before data modeling**
 This often improves the score. However, if those external data are injected at the final step, the score shows almost no improvement.

6 · References

[1] Z. Jiang *et al.*, AIDE: AI-driven exploration in the space of code, *arXiv:2502.13138*.
[2] E. Aygün *et al.*, An AI system to help scientists write expert-level empirical software, *Nature* **654**, 909 (2026).
[3] J. S. Chan *et al.*, MLE-bench: Evaluating machine learning agents on machine learning engineering, in *Proceedings of the 13th International Conference on Learning Representations*, 2025, https://openreview.net/forum?id=6s5uXNWGIh.
[4] NVIDIA Corporation, computer code NVIDIA Kaggle Plugin, Santa Clara, CA, 2026, https://github.com/NVIDIA/nvidia-kaggle.